

Fireworks for crossbow go through an entity if it is close to a player

MC-174214 COLLISION

Status	Resolved / Fixed
Affected	20w10a; 20w11a; 20w12a; 20w13b; 20w14a; 20w18a; 1.16 Release Candidate 1; 1.16; 1.16.1; 20w28a; 20w29a; 20w30a; 1.16.2 Pre-release 1; 1.16.2 Release Candidate 1; 1.16.2 Release Candidate 2; 1.16.2; 1.16.3; 1.16.4 Pre-release 1; 1.16.4 Release Candidate 1; 1.16.4; 20w45a; 20w51a; 21w03a; 1.16.5 (+more)
Fixed in	24w37a
Reported	2020-03-08
Mojira category	Collision; Projectiles

REPRODUCTION TARGET

Use version: 20w10a

Affected versions (113):

20w10a

20w11a

20w12a

20w13b

20w14a

20w18a

1.16 Release Candidate 1

1.16

1.16.1

20w28a

20w29a

20w30a

1.16.2 Pre-release 1

1.16.2 Release Candidate 1

1.16.2 Release Candidate 2

1.16.2

1.16.3

1.16.4 Pre-release 1

1.16.4 Release Candidate 1

1.16.4

20w45a

20w51a

21w03a

1.16.5

21w05a

21w05b

21w06a

21w07a

21w14a

21w15a

1.17 Pre-release 2

1.17 Pre-release 5

1.17 Release Candidate 1

1.17

1.17.1 Release Candidate 1

1.17.1

21w37a

21w40a

21w43a

1.18 Pre-release 5

1.18

1.18.1 Release Candidate 1

1.19

1.19.1 Release Candidate 2

1.19.1 Release Candidate 3

1.19.1

1.19.2 Release Candidate 1

1.19.2

22w42a

1.19.3 Release Candidate 1

1.19.3

23w04a

23w06a

1.19.4 Release Candidate 1

1.19.4 Release Candidate 3

1.19.4

23w16a

23w17a

23w18a

1.20 Pre-release 1

1.20 Pre-release 2

1.20 Pre-release 4

1.20 Pre-release 5

1.20 Pre-release 6

1.20 Pre-release 7

1.20 Release Candidate 1

1.20

1.20.1 Release Candidate 1

1.20.1

23w31a

23w32a

23w33a

23w35a

1.20.2 Pre-release 2

1.20.2 Release Candidate 1

1.20.2

23w40a

23w41a

23w42a

23w43b

23w44a

23w45a

23w46a

1.20.3 Pre-Release 2

1.20.3 Release Candidate 1

1.20.4

23w51b

24w03b

24w04a

24w05b

24w06a

24w07a

24w09a

24w11a

24w12a

24w13a

24w14a

1.20.5 Pre-Release 1

1.20.5 Pre-Release 3

1.20.5 Release Candidate 2

1.20.5

1.20.6 Release Candidate 1

1.20.6

24w18a

24w21b

1.21 Pre-Release 2

1.21 Pre-Release 4

1.21

1.21 Release Candidate 1

1.21.1

24w33a

24w34a

24w35a

Environment: Minecraft Java Edition snapshot 20w10a in a single-player world. Use Creative mode with commands enabled as a practical setup choice because the report requires a summon command but does not specify a game mode, seed, difficulty, multiplayer server, or other world settings.

PRECONDITIONS

- Launch the exact affected version 20w10a rather than the fix version 24w37a or a later version.
- Use a world in which the player can execute commands. A new Creative-mode world with commands enabled is the simplest setup.
- Perform the test in an open area with enough horizontal space to see, approach, and aim at a summoned enderman without nearby blocks obstructing the shot.
- Have one Crossbow, at least one Arrow, and at least one ordinary Firework Rocket available. The report does not specify custom firework NBT, colors, explosion stars, or flight duration.
- Know the configured controls for Use Item/Place Block, Open Chat, and Open Inventory. With default controls these are right-click, T, and E, respectively.
- Keep the offhand empty while loading the arrow control shot. Later, place the Firework Rocket in the offhand so the crossbow loads the rocket instead of another arrow.

STEPS TO REPRODUCE

1 REPORT

Launch Minecraft Java Edition using an installation configured for snapshot 20w10a. At the title screen, verify that the displayed version is 20w10a before continuing.

SUCCESS CUE The Minecraft title screen identifies the running game version as 20w10a.

2 INFERRED

Create or open a single-player Creative-mode test world with commands enabled. After joining, wait until the surrounding terrain has rendered and movement and inventory input respond normally.

INPUT

```
Singleplayer -> Create New World -> set Game Mode to Creative and ensure commands/cheats are allowed -> Create New World
```

SUCCESS CUE The player is in a responsive Creative-mode world and can use commands.

3 INFERRED

Move to an open, approximately level area. Open the Creative inventory, locate a Crossbow, an Arrow, and a Firework Rocket by their item names, and place them in accessible inventory or hotbar slots. Put the Crossbow in a known hotbar slot, keep the Arrow anywhere in the inventory, and keep the Firework Rocket out of the offhand for now.

INPUT

```
Press E; use the Creative inventory or Search Items interface to obtain Crossbow, Arrow, and Firework Rocket; press E again to close the inventory.
```

SUCCESS CUE A Crossbow is available in the hotbar, an Arrow is in the inventory, a Firework Rocket is available for later use, and the offhand is empty.

4 INFERRED

Select the Crossbow in the active hotbar slot. With the Arrow in the inventory and no Firework Rocket in the offhand, hold the Use Item control until the full crossbow-loading sequence and loading sound finish, then release the control.

INPUT

```
Press the hotbar number assigned to the Crossbow; hold right-click until loading completes; release right-click.
```

SUCCESS CUE The Crossbow remains in its visibly loaded state after right-click is released, with the arrow selected as its ammunition.

5 REPORT

While standing in the open test area, open in-game chat and execute the report's summon command exactly as written, including the leading slash and the historical NBT syntax. Do not modernize or otherwise alter the command.

INPUT

```
Press T, enter `/summon enderman ~ ~ ~ {NoAI:1b}` , then press Enter.
```

SUCCESS CUE A motionless enderman is present at or immediately beside the player's command-execution position.

6 **INFERRED**

Allow the summoned entity and nearby terrain to finish rendering. Because the command summons the enderman at the player's current coordinates, move backward only as much as necessary to stop intersecting it and to see its body clearly.

INPUT

Use S or another appropriate movement direction to move a short distance away, then release the movement control.

SUCCESS CUE The enderman is clearly visible, remains stationary because of NoAI, and can be approached without a block obstructing the line of fire.

7 **REPORT**

Face the enderman and move forward until the player is at point-blank range—as close to the enderman's collision area as movement and entity pushing permit. Aim the crosshair at the central torso rather than above the head, below the feet, or beside the body.

INPUT

Hold W while facing the enderman; release W when nearly touching it; adjust the mouse until the crosshair is centered on its torso.

SUCCESS CUE The enderman fills much of the forward view, the player is at the closest stable position next to it, and the crosshair points into its torso.

8 **REPORT**

Fire the already-loaded arrow from the crossbow once while maintaining the point-blank position and torso aim. A loaded crossbow fires on a single Use Item press; do not begin another loading cycle before this shot.

INPUT Right-click once.

SUCCESS CUE The arrow control shot has made contact, indicated by the normal entity impact response such as an impact sound or brief damage animation, and the enderman remains available for the rocket test.

9 **INFERRED**

Wait for the enderman's brief post-hit damage animation and damage-immunity interval to end before preparing the rocket shot. Do not repeatedly shoot it.

INPUT Wait approximately 1 second without attacking.

SUCCESS CUE The enderman is no longer displaying the immediate damage flash from the arrow shot and is ready for a separate projectile attempt.

10 **INFERRED**

Open the inventory and place the Firework Rocket in the shield-shaped offhand slot. Leave the Crossbow in its hotbar slot and leave the Arrow elsewhere in the inventory. Close the inventory after confirming the rocket is visibly held in the offhand.

INPUT

Press E; drag the Firework Rocket to the offhand slot; press E to close the inventory.

SUCCESS CUE The Firework Rocket is held in the player's offhand, and the unloaded Crossbow remains accessible in the hotbar.

11 **INFERRED**

Select the Crossbow again. While the Firework Rocket is in the offhand, hold the Use Item control until the full loading sequence finishes, then release it. Keeping the rocket in the offhand during the complete loading sequence ensures that this crossbow shot is loaded with the rocket rather than the inventory arrow.

INPUT

Press the Crossbow's hotbar number; hold right-click until loading completes; release right-click.

SUCCESS CUE The Crossbow remains visibly loaded after the completed loading cycle, and the Firework Rocket was in the offhand throughout that cycle.

12 **REPORT**

If the arrow impact displaced the enderman or the player moved while managing the inventory, again approach until the player is at the closest stable point-blank position permitted by collision or entity pushing.

INPUT Face the enderman, hold W until nearly touching it, then release W.

SUCCESS CUE The player has re-established point-blank range from the enderman.

13 **INFERRED**

Steady the view so the crosshair points through the central portion of the enderman's torso. Confirm that the loaded Crossbow is still selected and avoid aiming at empty space between the enderman's legs or beyond the edges of its narrow body.

INPUT

Move the mouse to center the crosshair on the enderman's torso; do not press right-click yet.

SUCCESS CUE A loaded crossbow is selected, the player is at point-blank range, and the crosshair is centered on the enderman's torso.

14 **REPORT** **TRIGGER**

Fire the firework-loaded crossbow exactly once while maintaining the point-blank position and torso aim. Hand control to the observation subsystem immediately as the shot is released; do not fire additional rockets or move away.

INPUT Right-click once.

SUCCESS CUE The loaded crossbow has just discharged its Firework Rocket toward the point-blank enderman, with no subsequent player action yet performed.

HANDOFF TO OBSERVATION

The player has just fired one Firework Rocket from the loaded crossbow while standing at point-blank range and aiming at the stationary enderman's torso. Begin observation immediately, before the player moves, reloads, or fires another projectile.

UNCERTAINTIES

- The report does not specify a game mode, seed, difficulty, or server type. A single-player Creative world with commands enabled is selected only to provide the required command access and items without adding combat, crafting, or exploration.
- The report describes the required distance only as 'very close' and gives no numeric threshold. The plan operationalizes this as the closest stable point-blank position permitted by entity collision or pushing.
- The report says only 'a fireworks rocket' and does not provide firework NBT, explosion-star contents, colors, or flight duration. The plan therefore uses an ordinary Firework Rocket obtained from the Creative inventory without custom data.
- The offhand loading procedure is an operational method for ensuring that the crossbow loads the Firework Rocket while arrows remain in the inventory; the report itself does not specify hand placement or inventory arrangement.
- The report uses the arrow shot as a control before the rocket shot. The short pause after the arrow impact is added to separate the two projectile attempts and avoid firing during the target's immediate post-hit interval.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Trigger reached?	YES / NO
Crash file	
Notes	
Tested by / date	